WEST BENGAL JEE 2026

CHEMISTRY MOCK TEST QUESTION

General Instructions

- This paper contains 75 questions divided into three categories.
- **Category-I:** 1 Mark each. Only one option correct. Negative marking: -0.25.
- **Category-II:** 2 Marks each. Only one option correct. Negative marking: -0.5.
- **Category-III:** 2 Marks each. One or more options may be correct. No negative marking.
- Partial marks are awarded for partially correct answers in Category-III.

Category: I (Q: 1 to 50)

1 Mark Each. Only one correct option.

1. In SO_2, NO_2^-, and N_3^-, the hybridizations at the central atom are respectively

(A) sp^2, sp^2 and sp

(B) sp^2, sp and sp

(C) sp^2, sp^2 and sp^2

(D) sp, sp^2 and sp

Ans: (A)

2. The type of hybridization and the magnetic property of $[MnCl_6]^{3-}$ are

(A) d^2sp^3, paramagnetic (4 unpaired e^-)

(B) sp^3d^2, paramagnetic (4 unpaired e^-)

(C) d^2sp^3, paramagnetic (2 unpaired e^-)

(D) sp^3d^2, paramagnetic (2 unpaired e^-)

Ans: (B)

3. Which of the following species is diamagnetic according to Molecular Orbital Theory?

(A) O_2

(B) B_2

(C) C_2

(D) N_2^+

Ans: (C)

4. The SI unit of molar conductivity is

(A) S cm^2 mol^{-1}

(B) S m^2 mol^{-1}

(C) cm^2 mol^{-1}

(D) Ω m mol^{-1}

Ans: (B)

5. The entropy of a perfect crystal at 0 K is

(A) Maximum

(B) Minimum but not zero

(C) Zero

(D) Infinite

Ans: (C)

6. The pH of pure water at 25°C is

(A) 0

(B) 14

(C) 7

(D) 1

Ans: (C)

7. Rate of diffusion of gases is governed by

(A) Boyle's law

(B) Charles' law

(C) Graham's law

(D) Dalton's law

Ans: (C)

8. The Bond Order of He2 is:

(A) 0

(B) 0.5

(C) 1

(D) 1.5

Ans: (B)

9. Which of the following molecules has a square pyramidal geometry?

(A) PCl_5

(B) BrF_5

(C) XeF_2

(D) SF_6

Ans: (B)

10. The increasing order of dipole moment for the following is:

(A) $CH_4 < NH_3 < H_2O$

(B) $H_2O < NH_3 < CH_4$

(C) $NH_3 < CH_4 < H_2O$

(D) $CH_4 < H_2O < NH_3$

Ans: (A)

11. A reaction whose rate is independent of concentration is of

(A) First order

(B) Second order

(C) Zero order

(D) Third order

Ans: (C)

12. For the cell: Mg | Mg^{2+} (1M) || Ag^+ (1M) | Ag. $E°(Mg^{2+}/Mg)=-2.37$ V, $E°(Ag^+/Ag)=+0.80$ V. The standard EMF is:

(A) 1.57 V

(B) 2.37 V

(C) 3.17 V

(D) -3.17 V

Ans: (C)

13. Which of the following is a state function?

(A) Heat

(B) Work

(C) Internal energy

(D) Path length

Ans: (C)

14. 22.4 L of an ideal gas at STP contains

(A) 6.022 x 10^23 molecules

(B) 1 gram

(C) 0.5 mole

(D) 2 moles

Ans: (A)

15. One mole of real gas (a=3.6, b=0.04, T=300K, V=10L, R=0.0821). The pressure is

(A) 2.36 atm

(B) 2.46 atm

(C) 2.56 atm

(D) 2.66 atm

Ans: (B)

16. Standard enthalpy of formation of an element in its standard state is

(A) 1

(B) Zero

(C) -1

(D) 100 kJ

Ans: (B)

17. A process is spontaneous if

(A) $\Delta H > 0$

(B) $\Delta S < 0$

(C) $\Delta G < 0$

(D) $\Delta G > 0$

Ans: (C)

18. Hybridization of carbon in CO2 is

(A) sp3

(B) sp2

(C) sp

(D) dsp2

Ans: (C)

19. SI unit of surface tension is

(A) N m-2

(B) N m-1

(C) J m-1

(D) Pa

Ans: (B)

20. The Arrhenius equation relates rate constant with

(A) Pressure

(B) Volume

(C) Temperature

(D) Concentration

Ans: (C)

21. Which molecule involves sp3d hybridization and has different bond lengths?

(A) XeF4

(B) BF3

(C) SF4

(D) CH4

Ans: (C)

22. According to VSEPR theory, the geometry of XeOF4 is:

(A) Octahedral

(B) Square Pyramidal

(C) Trigonal Bipyramidal

(D) Square Planar

Ans: (B)

23. Which gas shows maximum deviation from ideal behaviour at high pressure?

(A) H2

(B) He

(C) SO2

(D) N2

Ans: (C)

24. The atomic number of the element with configuration [Ne] $3s^2$ $3p^3$ is

(A) 14

(B) 15

(C) 16

(D) 13

Ans: (B)

25. Which of the following is diamagnetic?

(A) O_2

(B) B_2

(C) C_2

(D) NO

Ans: (C)

26. Which pair of ions has the same number of unpaired electrons?

(A) Ti^{3+} and V^{3+}

(B) V^{3+} and Ni^{2+}

(C) Fe^{3+} and Mn^{2+}

(D) Both B and C

Ans: (D)

27. The hybridization of Nickel in $[Ni(CN)_4]^{2-}$ is:

(A) sp^3

(B) dsp^2

(C) sp^3d^2

(D) d^2sp^3

Ans: (B)

28. The most electronegative element is

(A) Oxygen

(B) Nitrogen

(C) Fluorine

(D) Chlorine

Ans: (C)

29. Oxidation state of Cr in $K_2Cr_2O_7$ is

(A) +3

(B) +6

(C) +2

(D) +7

Ans: (B)

30. Which species has maximum number of unpaired electrons?

(A) O_2

(B) O_2^+

(C) O_2^-

(D) N_2

Ans: (A)

31. Shape of NH_3 molecule is

(A) Trigonal planar

(B) Tetrahedral

(C) Trigonal pyramidal

(D) Linear

Ans: (C)

32. Noble gas with lowest ionization energy is

(A) He

(B) Ne

(C) Ar

(D) Rn

Ans: (D)

33. Chief ore of aluminium is

(A) Hematite

(B) Galena

(C) Bauxite

(D) Magnetite

Ans: (C)

34. Which complex is most likely to be high spin?

(A) $[Fe(CN)_6]^{4-}$

(B) $[Fe(H_2O)_6]^{2+}$

(C) $[Co(NH_3)_6]^{3+}$

(D) $[Cr(CN)_6]^{3-}$

Ans: (B)

35. Central metal ion present in hemoglobin is

(A) Mg^{2+}

(B) Fe^{2+}

(C) Cu^{2+}

(D) Zn^{2+}

Ans: (B)

36. Electrolyte used in lead storage battery is

(A) HCI

(B) HNO_3

(C) H_2SO_4

(D) H_3PO_4

Ans: (C)

37. Highest first ionization energy in second period is of:

(A) F

(B) N

(C) O

(D) Ne

Ans: (D)

38. Coordination number of sulfur in SF_6 is:

(A) 4

(B) 5

(C) 6

(D) 2

Ans: (C)

39. The correct order of bond length is:

(A) $O_2^+ < O_2 < O_2^-$

(B) $O_2 < O_2^+ < O_2^-$

(C) $O_2^- < O_2 < O_2^+$

(D) $O_2 < O_2^- < O_2^+$

Ans: (A)

40. The correct order of electron affinity is

(A) F > CI > Br > I

(B) CI > F > Br > I

(C) CI > Br > F > I

(D) F > CI > I > Br

Ans: (B)

41. The standard electrode potential of a half-cell depends on:

(A) Nature of electrode

(B) Concentration of ions

(C) Temperature

(D) All of these

Ans: (D)

42. Hybridization of carbon in ethene is

(A) sp

(B) sp^2

(C) sp^3

(D) dsp^2

Ans: (B)

43. Alkene reacts with cold alkaline $KMnO_4$ to form

(A) Alcohol

(B) Ketone

(C) Vicinal diol

(D) Alkane

Ans: (C)

44. Which hydrocarbon burns with a sooty flame?

(A) Alkane

(B) Alkene

(C) Alkyne

(D) Aromatic hydrocarbon

Ans: (D)

45. Monomer of PVC is

(A) Ethene

(B) Vinyl chloride

(C) Styrene

(D) Tetrafluoroethylene

Ans: (B)

46. Reaction of alkyl halide with Na in dry ether gives alkane. This is called

(A) Fittig reaction

(B) Wurtz reaction

(C) Kolbe reaction

(D) Sandmeyer reaction

Ans: (B)

47. Which functional group undergoes nucleophilic addition?

(A) Alkene

(B) Alkane

(C) Carbonyl

(D) Ether

Ans: (C)

48. Which pair represents enantiomers?

(A) (R)-2-butanol & (S)-2-butanol

(B) cis-2-butene & trans-2-butene

(C) 1-butanol & 2-butanol

(D) meso-tartaric acid & d-tartaric acid

Ans: (A)

49. Glucose is a

(A) Protein

(B) Lipid

(C) Carbohydrate

(D) Vitamin

Ans: (C)

50. Polymer used in non-stick cookware is

(A) PVC

(B) Nylon

(C) Bakelite

(D) Teflon

Ans: (D)

Category: II (Q: 51 to 65)

2 Marks Each. Ground truth based on provided keys.

51. Which isomerism is shown by Diethyl ether and Methyl propyl ether?

(A) Chain isomerism

(B) Metamerism

(C) Functional isomerism

(D) Geometrical isomerism

Ans: (B)

52. Among the following, the extensive variables are

(A) V (Volume)

(B) P (Pressure)

(C) E (Internal energy)

(D) H (Enthalpy)

Ans: (A, C, D)

53. X is an extensive property, and x is an intensive property of a thermodynamic system. Which of the following statement(s) is (are) correct?

(A) xX is extensive

(B) x/X is intensive

(C) X/x is extensive

(D) dX/dX is intensive

Ans: (A, C)

54. For spontaneous polymerization, which of the following is (are) correct?

(A) ΔG is negative

(B) ΔH is negative

(C) ΔS is positive

(D) ΔS is negative

Ans: (A, B, D)

55. Which of the following compounds will show a positive result with Tollen's reagent?

(A) Glucose

(B) Fructose

(C) Benzaldehyde

(D) Acetone

Ans: (A, B, C)

56. Which of the following will undergo a Cannizzaro reaction when treated with concentrated NaOH?

(A) Formaldehyde

(B) Benzaldehyde

(C) Acetaldehyde

(D) Trimethyl acetaldehyde

Ans: (A, B, D)

57. Which of the following statements about SN^1 reactions are correct?

(A) It follows first-order kinetics

(B) It involves the formation of a carbocation intermediate

(C) It is favoured by polar protic solvents

(D) It results in complete inversion of configuration

Ans: (A, B, C)

58. Which of the following are examples of addition polymers?

(A) Polyethylene

(B) PVC

(C) Nylon-6,6

(D) Teflon

Ans: (A, B, D)

59. Which of the following compounds will give a yellow precipitate with Iodine and NaOH (Iodoform test)?

(A) Ethanol

(B) Acetophenone

(C) Benzophenone

(D) Isopropyl alcohol

Ans: (A, B, D)

60. Which of the following reagents can be used to distinguish between Ethene and Ethyne?

(A) Bromine water

(B) Ammoniacal Cuprous Chloride

(C) Ammoniacal Silver Nitrate

(D) Alkaline KMnO4

Ans: (B, C)

61. Which of the following are periodic properties that generally decrease down a group in the periodic table?

(A) Atomic radius

(B) Electronegativity

(C) Ionization enthalpy

(D) Electron gain enthalpy (less negative)

Ans: (B, C, D)

62. Which of the following undergo nucleophilic substitution (SN1) faster than Ethyl Chloride?

(A) Benzyl Chloride

(B) Tert-butyl Chloride

(C) Allyl Chloride

(D) Chlorobenzene

Ans: (A, B, C)

63. Which of the following are examples of intensive properties?

(A) Temperature

(B) Density

(C) Molar Heat Capacity

(D) Volume

Ans: (A, B, C)

64. Which of the following properties are characteristic of transition elements?

(A) Formation of coloured ions

(B) Variable oxidation states

(C) Paramagnetic behaviour

(D) Formation of interstitial compounds

Ans: (A, B, C, D)

65. Which of the following compounds are aromatic in nature?

(A) Benzene

(B) Naphthalene

(C) Anthracene

(D) Cyclooctatetraene

Ans: (A, B, C)

Category: III (Q: 66 to 75)

2 Marks Each. Multiple correct options.

66. In which of the following reactions is the major product formed according to Saytzeff's rule?

(A) Dehydration of 2-Butanol

(B) Dehydrohalogenation of 2-Bromobutane with Alc. KOH

(C) Dehydrohalogenation of 2-Bromobutane with t-BuOK

(D) Pyrolysis of quaternary ammonium hydroxides

Ans: (A, B)

67. Which of the following statements about Enzymes are correct?

(A) Highly specific in nature

(B) Proteinaceous in nature

(C) Lose activity at very high temperatures

(D) Act as biocatalysts

Ans: (A, B, C, D)

68. Which of the following properties of Alkali metals increase as the atomic number increases?

(A) Atomic radius

(B) Ionic radius

(C) Density (exception of K)

(D) First Ionization Enthalpy

Ans: (A, B, C)

69. Which of the following statements about the 'Inert Pair Effect' are true?

(A) Prominent in heavier p-block elements

(B) Reluctance of s-electrons to bond

(C) Stability of lower oxidation states in Gr 14/15

(D) Caused by poor shielding of d and f orbitals

Ans: (A, B, C, D)

70. Which of the following statements regarding $KMnO_4$ are correct?

(A) Oxidizing agent in all media

(B) Reduced to Mn^{2+} in acidic medium

(C) Color due to d-d transitions

(D) Volumetric reagent in redox titrations

Ans: (A, B, D)

71. Which of the following statements are correct for the Group 15 elements (Nitrogen family)?

(A) Stability of +3 increases down group

(B) Stability of +5 decreases down group

(C) Nitrogen can form pπ-pπ multiple bonds

(D) Bismuth is strong oxidizing agent in +5

Ans: (A, B, C, D)

72. Which compound shows geometrical isomerism?

(A) $CH_2=CH_2$

(B) CH_3-CH_3

(C) $CH_3-CH=CH-CH_3$

(D) $CH_3-C\equiv C-CH_3$

Ans: (C)

73. The IUPAC name of $CH_3-CH(OH)-CH_3$ is

(A) Propan-1-ol

(B) Propan-2-ol

(C) Isopropyl alcohol

(D) Ethanol

Ans: (B)

74. Which reaction forms an ether from an alkyl halide?

(A) Wurtz reaction

(B) Cannizzaro reaction

(C) Williamson synthesis

(D) Kolbe reaction

Ans: (C)

75. The configuration of a chiral carbon is determined using:

(A) Huckel rule

(B) CIP priority rules

(C) VSEPR theory

(D) MOT

Ans: (B)